From Correlation to Causation: Differential Causal Networks

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1 Background

Rewiring of gene interactions under different conditions is a fundamental cause of different phenotypic responses [1]. Differential network analysis, among others, is an effective tool for investigating gene and protein network rewiring. There exist many computational methods to evaluate differential networks discriminating against these conditions. More recently, motivated by the work by Judea Pearl [2], the relevance of the determination of causal effects among molecules has known an increased interest by the research community. Despite this, there is a lack of determination of differences in causal effects between two conditions as a counterpart of differential network analysis. Therefore, we present here a prototype strategy to determine differential causal networks between two conditions, focusing on sex as a differential factor.

2 Methods

We start from gene expression data taken from GTEx database [3] from some selected tissues, focusing on genes correlated to Type 2 Mellitus Diabetes, subdivided by sex. We explored causal effects considering the age evolution by using the DoWhy Python Package. Moreover, we made use of pcalg Python Package to find an illustration of the relationships among some of the considered genes.

2.1 Causal Inference: We conducted our initial analyses on nine tissues (adipose tissue, amygdala tissue, aorta tissue, blood tissue, cerebellum tissue, colon tissue, coronary tissue, liver tissue, and lung tissue) to investigate the potential effects of the considered genes on age progression, treating genes as interventions and age as the outcome. In our preliminary results, we identified causal effects performed on pharmacologically approved genes extracted from the GTEx database for patients with Type 2 Diabetes Mellitus. However, we filtered out genes that were not expressed in the majority of instances. For each causal inference-based analyses, we computed a refutation test based on random causes, and the newly obtained effects are closely aligned with the estimated effects. According to this, age progression varies significantly between sexes and different tissues due to the gene expression.

2.2 Causal Discovery: The second phase of our work was based on applying causal discovery to the same data obtained from the GTEx database, but with the addition of a gene filter: we eliminated genes whose average expression showed low levels. This analyses aimed to investigate the differences in causal graph structures between sexes. We utilized the PC algorithm [5] as a causal discovery method to identify the causal structures (two for each tissue, one for each sex). Subsequently, we used some NetworkX functions to identify differences in the edges of causal networks. In all the analyses, there are some links that differ between the two causal networks of men and women. This fact proves that there are different causal mechanisms in Type 2 Mellitus diabetes between sexes.

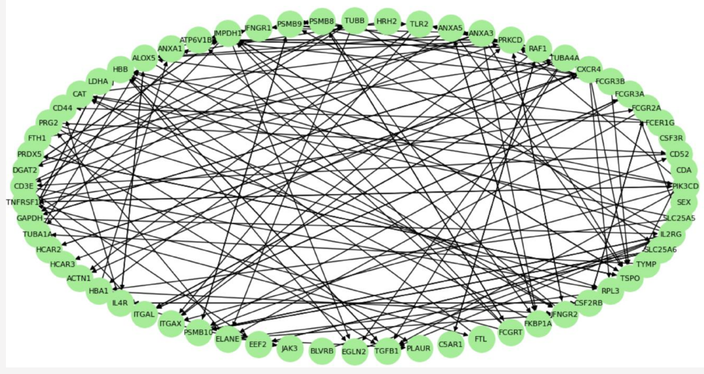

Figure 1: Total differential Network

For example, in the case of blood tissue, there are 155 edges of difference between the causal networks: the men's network contains 82 edges that women's network doesn't contain, the women's network contains 73 edges that men's network doesn't contain.

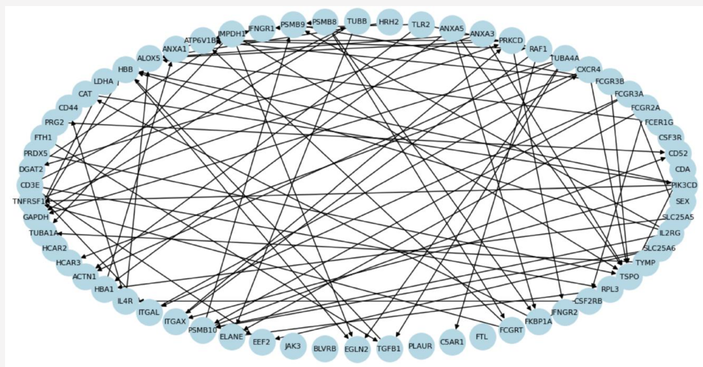

Figure 2: Differential Network showing which edges exist in males but not in females

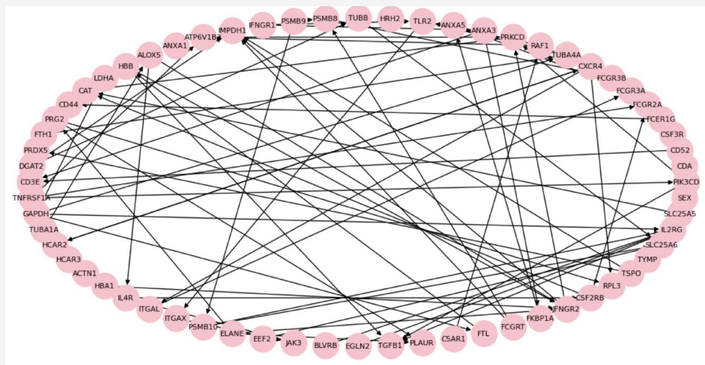

Figure 3: Differential Network showing which edges exist in females but not in males

3 Conclusions

Our results indicate that there are causal effects on age evolution considering genes as treatments. Furthermore, causal discovery demonstrates different causal mechanisms in Type 2 Mellitus diabetes between sexes considering these genes that are active in response to drugs.

4 References

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